

1 Three Dimensional Coordinate Systems and Vectors

1.2 Vectors

It will be convenient to use vectors along side coordinate systems. If we use our x , y , and z -axes from before, and we place a vector $\vec{v}$ at the origin, the terminal end of the vector has coordinates which we will use to describe the vector itself.

Each vector has a length associated to it, this can be calculated by the following formula.

Definition 1.2.1 The length of a vector $\vec{v} = \langle a_1, a_2 \rangle$ is obtained by the formula

$$\sqrt{a_1^2 + a_2^2},$$

and is denoted $|\vec{v}|$. Similarly, for a 3-dimensional vector $\vec{u} = \langle a_1, a_2, a_3 \rangle$, the length is given by

$$|\vec{u}| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

When using vectors with a coordinate system, we have a convenient way of describing the vector operations of addition and scalar multiplication:

Definition 1.2.2 If $\vec{v} = \langle a_1, a_2, a_3 \rangle$, and $\vec{u} = \langle b_1, b_2, b_3 \rangle$, then

$$\vec{v} + \vec{u} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle,$$

and if c is a scalar, then

$$c\vec{v} = \langle ca_1, ca_2, ca_3 \rangle.$$

Theorem 1.2.3 *if $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors, and c and d are scalars, then the following properties hold:*

1. $\vec{a} + \vec{b} = \vec{b} + \vec{a}$.
2. $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$
3. $\vec{a} + \vec{0} = \vec{a}$
4. $\vec{a} + (-\vec{a}) = \vec{0}$
5. $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$
6. $(c + d)\vec{a} = c\vec{a} + d\vec{a}$
7. $(cd)\vec{a} = c(d\vec{a})$
8. $1\vec{a} = \vec{a}$.

Example 1.2.4 If $\vec{a} = \langle 4, 0, 3 \rangle$ and $\vec{b} = \langle -2, 1, 5 \rangle$, find $a + b$, $a - 2b$, $3a$, and $|a|$.

For 3-dimensional space, there are 3 vectors which play a special role, namely

$$\vec{i} = \langle 1, 0, 0 \rangle, \quad \vec{j} = \langle 0, 1, 0 \rangle, \quad \vec{k} = \langle 0, 0, 1 \rangle.$$

These are called the standard basis vectors, and we will often express vectors in terms of $\vec{i}$, $\vec{j}$, and $\vec{k}$. For example, we have that if $\vec{a} = \langle 3, -2, 1 \rangle$, then

$$\vec{a} = \langle 3, -2, 1 \rangle = 3\vec{i} - 2\vec{j} + \vec{k}.$$

Example 1.2.5 For the vectors $\vec{a} = \vec{j} - \vec{k}$ and $\vec{b} = 3\vec{i} + 2\vec{j}$, find $\vec{a} + 2\vec{b}$.

Definition 1.2.6 A unit vector is a vector whose length is 1.

The vectors $\vec{i}$, $\vec{j}$ and $\vec{k}$ are all unit vectors. If we have a vector $\vec{a}$, and we wish to find a unit vector in the same direction as $\vec{a}$, we compute

$$\vec{u} = \frac{\vec{a}}{|\vec{a}|}.$$

Example 1.2.7 Find a unit vector in the direction of $\langle 2, 3, -1 \rangle$.

Example 1.2.8 A 100-lb weight hangs from two wires such that the interior angles of the wires are 50° and 32° . Find the tension in the two wires and the magnitudes of these tensions.

